

AI-Powered Eldercare Platform

PulseCare

AI-Powered Senior Health Monitoring

February Jiang · yj2918

"Your body has a story. Pulse remembers it."



Motivation & Positioning

72M seniors hide pain from families — crises arrive without warning.

Five years ago, I lost my grandmother.

In the time leading up to her passing, she experienced a lot of physical discomfort—but she rarely spoke about it. Like so many seniors, she systematically hid her symptoms simply because she didn't want to worry our family. She hid her physical pain to avoid worrying our family. ***That silence — and the fall that followed — is why PulseCare exists.***

Seniors Under-Report

72M independent US seniors systematically hide symptoms to avoid burdening family. Disclosure only happens when pain is severe enough to admit — often too late.

Private Family Doctor

Our Positioning: An always-on "private doctor" for the family, providing early warnings preventative care and most importantly, actionable, professional medical guidance.

Shared Family Bond

It transforms care from a silent worry into a shared family bond—ensuring that our parents never face aging alone, but feel constantly supported and accompanied by the whole family.

Business Model & Technical Building

High-margin SaaS powered by conditional multi-agent AI — cost scales sub-linearly with users.

Unit Economics

98.8%

Gross Margin

\$29.00 /mo

Subscription per caregiver

~\$0.06

LLM cost (Gemini 2.0 Flash, conditional routing)

~\$0.36

Total COGS (infra + compute)

\$28.64

Contribution margin per user

Moat: after 60 days the senior's longitudinal baseline is the product — irreplaceable clinical memory.

Conditional Multi-Agent Routing

LangGraph: Companion Agent handles every check-in cheaply (~\$0.001). AnalystAgent fires only when severity ≥ 5 or symptoms are detected (~\$0.015). A 'I'm fine' costs nothing meaningful.

Cross-Modal Synthesis

AnalystAgent fuses 60-day conversation history with Apple Watch biometrics (HR, SpO₂, HRV, sleep stages). Divergence between reported status and vitals is itself the alert — e.g. 'I feel fine' + SpO₂ 93%.

Medical RAG Pipeline (FAISS + BM25)

Responses grounded in curated geriatric knowledge via Reciprocal Rank Fusion. Eliminates medical hallucinations and substantiates the \$29/mo clinical trust premium.

Structured Tool Calling

log_health_entry is force-called before every warm reply — guaranteeing qualitative conversation converts into structured DB rows powering every downstream alert. Enforces automated database logging from natural language inputs for dashboard visibility.